

Pathogenesis of type 2 diabetes: tracing the reverse route from cure to cause.

The metabolic abnormalities of type 2 diabetes can be reversed reproducibly by bariatric surgery. By quantifying the major pathophysiological abnormalities in insulin secretion and insulin action after surgery, the sequence of events leading to restoration of normal metabolism can be defined. Liver fat levels fall within days and normal hepatic insulin sensitivity is restored. Simultaneously, plasma glucose levels return towards normal. Insulin sensitivity of muscle remains abnormal, at least over the weeks and months after bariatric surgery. The effect of the surgery is explicable solely in terms of energy restriction. By combining this information with prospective observation of the changes immediately preceding the onset of type 2 diabetes, a clear picture emerges. Insulin resistance in muscle, caused by inherited and environmental factors, facilitates the development of fatty liver during positive energy balance. Once established, the increased insulin secretion required to maintain plasma glucose levels will further increase liver fat deposition. Fatty liver causes resistance to insulin suppression of hepatic glucose output as well as raised plasma triacylglycerol. Exposure of beta cells to increased levels of fatty acids, derived from circulating and locally deposited triacylglycerol, suppresses glucose-mediated insulin secretion. This is reversible initially, but eventually becomes permanent. The essential time sequence of the pathogenesis of type 2 diabetes is now evident. Muscle insulin resistance determines the rate at which fatty liver progresses, and ectopic fat deposition in liver and islet underlies the related dynamic defects of hepatic insulin resistance and beta cell dysfunction. These defects are capable of dramatic reversal under hypoenergetic feeding conditions, completely in early diabetes and to a worthwhile extent in more established disease.